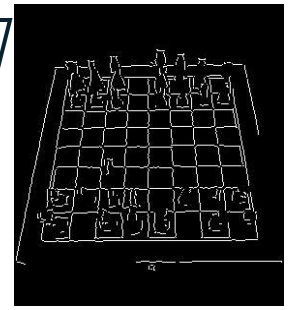
Image Based Mobile Phone Application for Chess Position Recognition

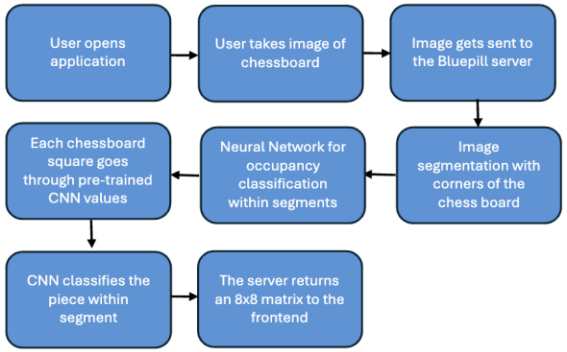
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Edward Delp | Carla Zoltowski | Maggie Zhu
College of Engineering, Purdue University | West Lafayette, IN

Abstract

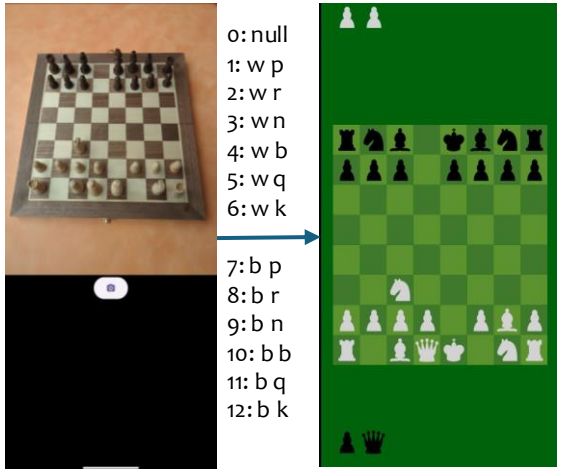
Studying an over-the-board chess game with engine analysis requires the tedious task of manually inputting a chess board configuration online. Our project seeks to eliminate this problem by making an Android application for chess players to configure the pieces with only an image of the chessboard.



Application Workflow



Flutter Application



Methodology

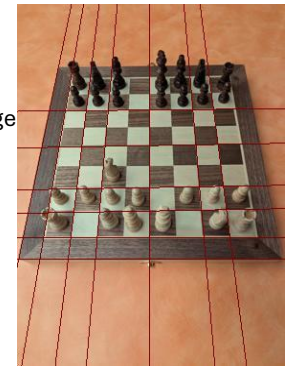
Board Localization

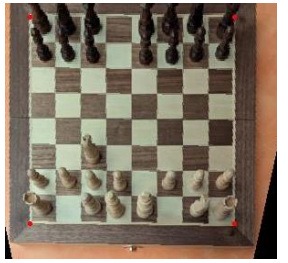
To process the image quickly, the image is first converted to grayscale and resized, such that each segmented square has side length 28. The Canny Operator is then applied to find the edges, optimizing noise reduction, gradient calculation, and edge tracking with hysteresis for precise and continuous edge mapping. With these edges, we run a Hough Line transform to sum the non-zero-pixel intensities in an accumulator matrix (1):

1) $\rho = x\cos(\theta) + y\sin(\theta)$

2) $\sqrt{(x-x')^2 + (y-y')^2} < \gamma.$

Lines are then grouped using their intersections as a heuristic. To account for both missing and extra lines, a RANSAC-based algorithm is employed to determine the homography matrix that best fits the largest number of inliers on the chessboard within a certain tolerance as shown above right (2). From the set of inliers, it is possible to identify whether lines are missing on the left, right, top, or bottom of the chessboard. Using the computed homography matrix, the previous Canny edge-detected image is warped to locate the missing horizontal and vertical lines, ultimately identifying the corners of the board.





Occupancy Classification

The neural network features an input layer with 784 neurons, corresponding to the 28x28 input dimensions, and an output layer with 2 neurons for binary classification. It includes a single hidden layer comprising 30 neurons, which captures intricate patterns within the data. Stochastic gradient descent (SGD) with backpropagation is employed to optimize weights and biases, iteratively minimizing classification errors. A quadratic cost function quantifies prediction discrepancies. Evaluation on the test set reveals exceptional performance, with accuracy exceeding 99%, showcasing the network's effectiveness in occupancy classification.

$$\delta^{(l)} = \left(W^{(l+1)}\right)^T \cdot \delta^{(l+1)} \circ \sigma'\left(z^{(l)}\right) \qquad \delta^{(L)} = \nabla_a C \circ \sigma'\left(z^{(L)}\right)$$

Piece Classification

The Convolutional Neural Network classify chess pieces by extracting spatial features through convolutional layers, reducing dimensions with pooling layers, and utilizing fully connected layers for classification. The network uses a loss function, cross-entropy, to quantify prediction errors and updates weights through backpropagation, optimizing feature learning for accurate piece recognition. The model uses 5 epochs with a batch size of 32 and a learning rate of 0.01.

